

AeroCrypt

Secure UAV Telemetry System Using AES-128-GCM and 433 MHz LoRa

Professional Project Report

An embedded UAV telemetry and ground-monitoring platform combining a 250-size FPV-style quadcopter, 433 MHz SX1278 LoRa communication, AES-128-GCM authenticated encryption, an ESP32 ground receiver, and a real-time Streamlit monitoring dashboard.

Project Domain	Embedded Systems • UAV • Wireless Communication • Cybersecurity
Security	AES-128-GCM authenticated encryption
Wireless	SX1278 LoRa, 433 MHz
Ground Receiver	ESP32 + SX1278
Monitoring	Streamlit real-time telemetry dashboard
UAV Platform	250-size ZMR250/QAV250 quadcopter

Prepared for technical documentation, portfolio presentation, and GitHub publication.

AeroCrypt — UAV Hardware Reference

250-size FPV-style quadcopter build bench and flight hardware



Figure. AeroCrypt UAV hardware reference showing the assembled 250-size quadcopter, motors, ESCs, flight-control electronics, radio equipment, 3S 1500 mAh 40C LiPo battery, and associated FPV/build components.

1. Executive Summary

AeroCrypt is a secure UAV telemetry project designed to protect wireless flight and sensor information while providing a practical ground-station interface for monitoring. The system combines a 250-size quadrotor platform with a 433 MHz SX1278 LoRa link and AES-128-GCM authenticated encryption. On the ground side, an ESP32 receives LoRa packets and provides the recovered telemetry to a Streamlit dashboard for live visualization.

The project emphasizes the complete telemetry path: telemetry preparation, authenticated encryption, LoRa transmission, ground reception, authentication/decryption, and operator-facing visualization. The dashboard provides visibility into environmental measurements, inertial measurements, packet information, LoRa link quality, GPS status, and security verification.

2. Project Objectives

- Develop a long-range wireless telemetry link suitable for a small UAV platform.
- Protect telemetry confidentiality and integrity using AES-128-GCM.
- Use SX1278 LoRa modules at 433 MHz for wireless telemetry.
- Implement an ESP32-based ground receiver.
- Provide a real-time Streamlit monitoring interface.
- Expose practical indicators including RSSI, SNR, packet count, and authentication status.
- Maintain a lightweight 250-size quadcopter configuration for FPV-style operation.

3. System Architecture

End-to-end flow:

UAV telemetry → AES-128-GCM protection → SX1278 LoRa transmitter → 433 MHz wireless link → SX1278 receiver → ESP32 ground receiver → USB serial → Streamlit dashboard.

Stage	Function
UAV / telemetry source	Provides flight, environmental, inertial, and communication-related telemetry.
Security layer	Applies AES-128-GCM authenticated encryption before wireless transmission.
LoRa transmitter	Transfers protected telemetry over the 433 MHz link.
LoRa receiver	Receives the wireless packet at the ground station.
ESP32 ground receiver	Processes received packets and exposes telemetry to the host computer.
Streamlit dashboard	Displays live telemetry, history, link quality, and security state.

4. UAV Hardware Configuration

The intended aircraft is a 250-size FPV-style quadcopter built around a ZMR250/QAV250 carbon-fiber frame. The configuration is intended to balance stable flight, manageable weight, and reasonable flight time.

Component	Configuration
Frame	ZMR250 / QAV250 carbon-fiber 250-size frame
Motors	4 × 2200 KV generic brushless motors
ESCs	4 × 30 A ESCs
Flight controller	KK2.1.0
Radio control	Basic drone transmitter and receiver
Telemetry	SX1278 LoRa, 433 MHz

5. Battery and Power System

The proposed initial flight battery is a 3S LiPo pack. A 3S 1500 mAh 40C battery is the planned starting configuration. The stated 4–6 minute flight time is an estimate and depends on final aircraft weight, propeller selection, battery condition, and flying style. A laptop charger or ordinary 12 V adapter is not a substitute for the high-current flight battery.

Parameter	Target configuration
Battery type	LiPo
Configuration	3S
Nominal voltage	11.1 V
Fully charged	12.6 V
Capacity	1500–1800 mAh
Discharge rating	35C–50C or higher
Connector	XT60
Initial choice	3S 1500 mAh 40C
Expected flight time	Approximately 4–6 minutes; dependent on final build and flight conditions

6. AES-128-GCM Security Layer

AeroCrypt uses AES-128-GCM as the telemetry security mechanism. GCM provides encryption together with authentication, allowing the receiver to verify a protected packet before accepting recovered telemetry.

Security sequence: Telemetry → AES-128-GCM encryption → protected LoRa packet → reception → GCM authentication → decryption / recovery → dashboard.

The supplied dashboard evidence includes an explicit **AES-128-GCM VERIFIED** state indicating that the receiver successfully authenticated the latest decrypted packet in the demonstrated session.

Public-release rule: real encryption keys, Wi-Fi credentials, broker credentials, private keys, and other secrets must never be committed to a public repository. Public configuration should use placeholders.

7. Telemetry and Link Monitoring

The dashboard presents sensor, inertial, packet, and radio-link information. The supplied evidence demonstrates live packet reception and historical visualization.

Category	Displayed information
Environmental	Temperature and pressure
Accelerometer	X, Y, and Z acceleration
Gyroscope	X, Y, and Z angular-rate values
LoRa link	RSSI and SNR
Packet	Packet number, time, and received-packet count
GPS	GPS status; location/satellite fields when telemetry is available

8. Streamlit Ground Monitoring Dashboard

The Streamlit dashboard acts as the operator-facing ground station. It provides serial configuration, live receiver status, packet information, telemetry cards, historical plots, a telemetry table, GPS state, LoRa link indicators, and security verification.

- Serial-port and baud-rate configuration.
- Live receiver connection status.
- LoRa packet count and latest-packet display.
- Temperature and pressure history plots.
- Accelerometer and gyroscope history plots.
- Recent telemetry table.
- RSSI and SNR monitoring.
- GPS waiting/availability status.
- AES-128-GCM verification status.
- Telemetry history and CSV-export controls.

9. Project Evidence and Interface Screenshots

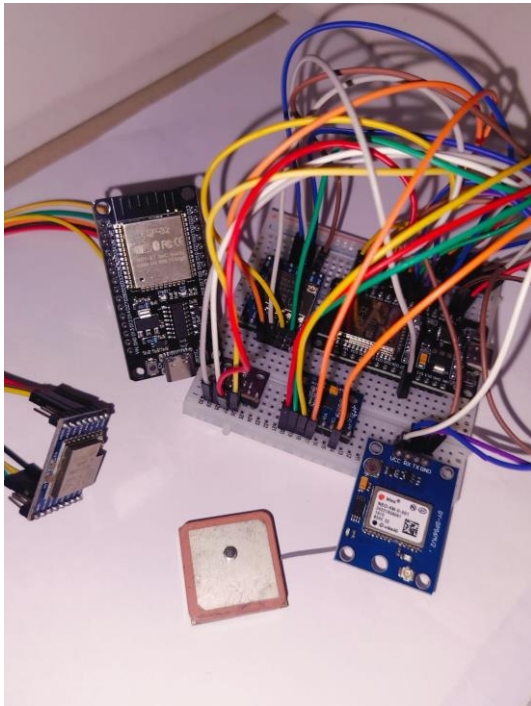


Figure 1. Physical UAV/electronics hardware and telemetry-related components.

Figure 2. Streamlit ground-station overview showing receiver, LoRa, encryption, and telemetry status.

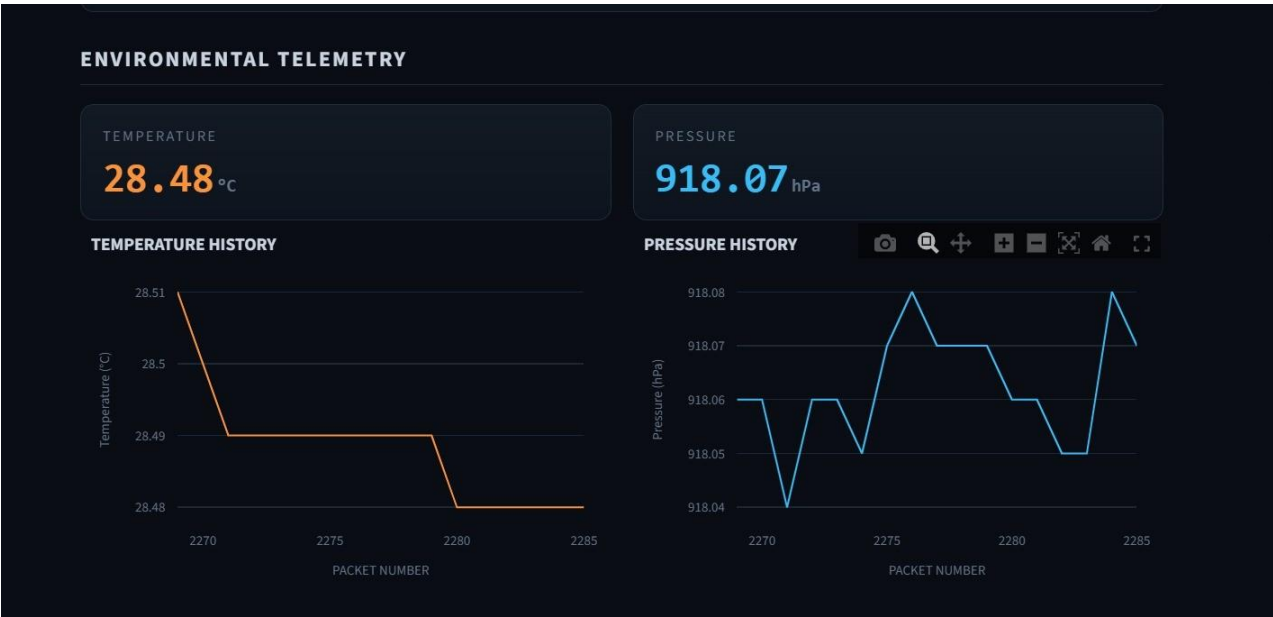


Figure 3. Environmental telemetry: temperature and pressure history.



Figure 4. MPU accelerometer telemetry and packet history.



Figure 5. MPU gyroscope telemetry and packet history.



Figure 6. Telemetry table with recent packets and link-quality information.

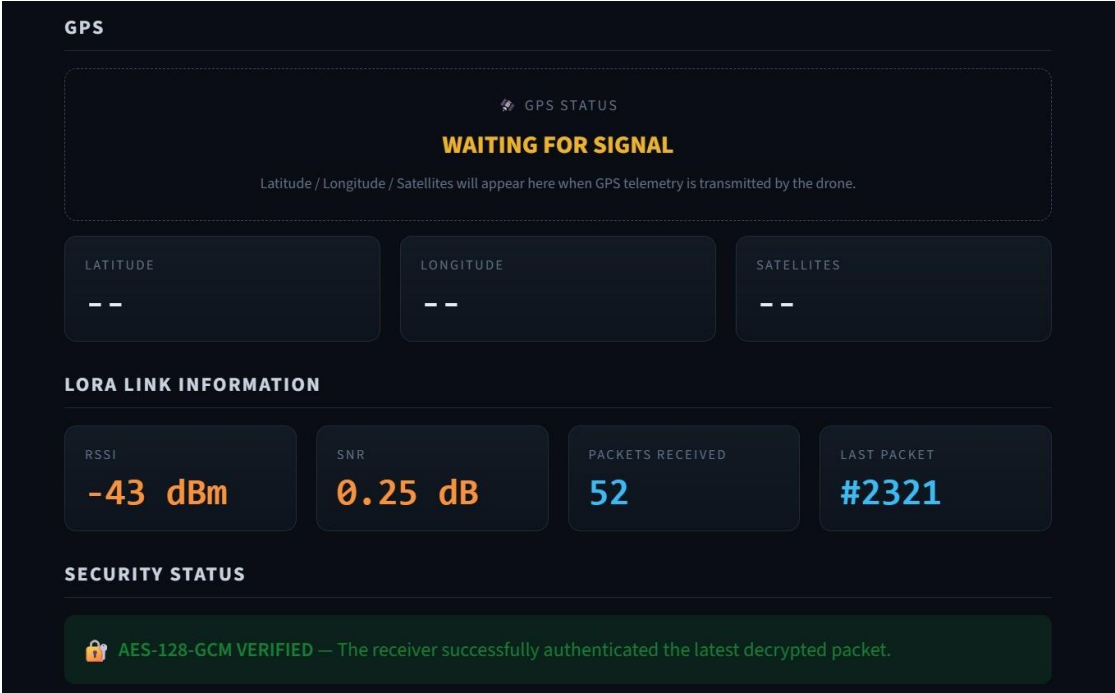


Figure 7. GPS and LoRa link information with AES-128-GCM verification status.

10. Software and Implementation

The project combines embedded receiver firmware with a Python Streamlit ground-station interface. The public repository should contain the intended source, configuration templates, documentation, and sample material that can safely be released.

Layer	Technology / role
Embedded platform	ESP32
LoRa transceiver	SX1278, 433 MHz
Security	AES-128-GCM
Ground application	Python + Streamlit
Host connection	USB serial
Visualization	Live telemetry cards, plots, tables, and link indicators

11. Demonstrated Results

The supplied dashboard evidence demonstrates an operational telemetry session with received-packet counts, live sensor values, accelerometer and gyroscope traces, RSSI/SNR information, a telemetry table, and an AES-128-GCM verification state. Example displayed values include approximately 28.48 °C temperature, 918.06 hPa pressure, RSSI values around -40 to -45 dBm, and active packet reception. These are observed dashboard-session values and should not be treated as universal performance specifications.

No fixed packet-delivery percentage, latency benchmark, or flight-endurance measurement is claimed here because such metrics require a defined experimental procedure and repeated measurements.

12. Safety and Engineering Considerations

A 250-size quadcopter contains rotating propellers, high-current ESCs, and a high-discharge LiPo battery. Battery handling, charging, wiring, motor testing, and flight operations should follow appropriate safety procedures. Propellers should be removed during bench testing of motors and control electronics.

The flight battery is a high-current power source and must be selected and operated according to the motor, ESC, propeller, current, and final aircraft-weight requirements.

13. Current Scope and Limitations

The documented AeroCrypt scope is secure LoRa telemetry and ground monitoring. This report does not claim autonomous flight, advanced attitude estimation, machine-learning intrusion detection, GPS mapping, or formal radio-performance benchmarks unless those features are separately implemented and experimentally validated.

14. Future Enhancements

- Integrate telemetry transmission more tightly with the final flight-control and sensor stack.
- Add systematic range, packet-loss, latency, and power-consumption experiments.
- Add persistent telemetry storage and reproducible data-export workflows.
- Add stronger packet framing and replay-protection measures where required by the final protocol.
- Add controlled GPS visualization after validating GPS telemetry availability.
- Perform structured flight testing to characterize endurance under defined payload and propeller conditions.

15. Conclusion

AeroCrypt demonstrates an end-to-end approach to secure UAV telemetry by combining a 250-size quadcopter platform, 433 MHz LoRa communication, AES-128-GCM authenticated encryption, an ESP32 ground receiver, and a real-time Streamlit monitoring interface. The architecture provides a practical foundation for secure wireless telemetry experimentation in embedded and UAV systems while remaining modular for future flight, communication, and security work.

Public-release note: Before publishing AeroCrypt to GitHub, remove all real credentials and cryptographic keys, use placeholders in public configuration, and include only source and documentation intended for public release.